

Transformer Laminations



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Mike LaFevre (of Magnaquest fame) posted this article to rec.audio.tubes, in response to some questions from Fred Ireson. It gives a great insight into the different types of transformer laminations available, and the reasons for selecting them.

Here's the whole article, reproduced with Mikes permission.

From: acrosound@aol.com (Acrosound)

Newsgroups: rec.audio.tubes

Subject: Re: M4 vs.M6 vs. M19 for Mikey ??

Message-ID: <1998091306582800.CAA02403@ladder01.news.aol.com>

Fred Ireson wrote:

>Dear Mike - Lets say you design and build three output transformers
>with identical (close as possible due to core size difference)
>specifications, all having the same primary inductance, turns ratio,
>power handling, and interleaving but they are made using these three
>grades of iron.

May I propose that we do our "model" just a bit differently. Instead of varying the core size (which I would understand as core area) let us keep it the same. At least for a moment.

If we wound three identical coils (within real world "identicalness") and stacked them with M4, M6, and M19 core materials what might we expect to be the differences?

One of the differences would be the amount of inductance produced. Again, to simplify our model, let us assume a PP trans and no DC imbalance. Then the material with the highest perm will produce the most inductance. Here the pecking order would be M4, M6, M19 in descending order.

Another difference btwn the materials would be the amount of losses. This is generally stated by manufacturers of the electrical grades of steels as a watts per pound at a specific flux level. Usually they will provide a graph that shows the relationship btwn flux level and losses. the higher the flux level the higher the losses. Again, M4 will exhibit the least amount of losses and M19 the most losses.

With the higher losses we might also expect higher temperature rises.

You mention "interleaving".... let us assume in the case above that the interleaving is the same... say 1X1. Interleaving opens up a lot of interesting discussions. Partridge in his seminal papers written in the late thirties explored "interleaving" in pretty great depth. Ercel Harrison wrote a paper on why he preferred interleaving 2x2 in many cases. But this would take us far afield at this time.

Now go back to your original idea. Vary the core size to produce a high degree of constancy (same inductance, power handling, and turns ratios) in the three different samples. One expected outcome would be that the M4 and M6 cores would be smaller than the M19 core size under this scenario. All other things being equal we would want the core to be as small as is possible (while still meeting our performance standards). Why? The smaller core will have a smaller mean length of turn (copper circuit). And this is desirable since we want to minimize capacity....the smaller the plate areas (think of any two succeeding layers of windings as being the discrete plates of a capacitor), again, all other things being equal (in this case the spacing btwn the plates) the less capacitance you will have. Also our copper losses may be easier to keep smaller in magnitude on the smaller core (but we would need to do complete designs to prove this out and to optimize for this effect if desired).

In practice the M4 and M6 cores would wind up being smaller in size than the M19 outfitted trans if we wanted to try to acheive the desired goal that you state in the above. And for clarification....let us assume use only of the conventional shaped EI lamination shapes for now.

Let me note another important difference btwn the M4, M6 and M19 products. The gauge that these are provided in are different in each case (again, assuming we restrict ourselves for the moment to the conventional EI shapes).

The M19 is available in 26 (.0185") and 24 gauge. The M6 is (mostly) available in 29 gauge (.014"). The M4 is made .011" thick (I forget what guage that works out to be).

>Would you briefly describe the objective and subjective differences
>between these? (I suspect in the "Real World" of RAT that the
>interleaving attention couldn't be applied to the M19 version due to
>economics)

M6 is the conventional standard bearer of quality amongst the grades you listed. It is grain oriented (the M4 is as well) while the M19 is not. Somewhere in old literature that I have I saw the M19 referred to as "power transformer" grade material. M19 can not be run at as high a flux density as M6 or M4 grades and will exhibit higher amounts of magnetically induced core distortions, require higher magnetizing current, and have greater losses. It is a less efficient material than M6 or M4 electrically.

Cost wise m19 costs about 15 to 20 percent less than M6 grade. M4 grade, on the other hand, costs about double (200%) more than M6.

Subjective differences? Much tougher to explain.... let me think this one through a bit..... there are I believe "subjective" differences btwn these grades with the materials exhibiting the moreso favorable electrical properties also being the preffered (in my sonic orbits) subjective material. But again, bear in mind, that core materials is only one piece of the puzzle.. albeit a very large piece of the puzzle.... so that as I stated in another thread a good design is never (in my experience) reducible to only one factor or consideration.

Also in single ended designs.... because of their uniqueness electrically and magnetically the same order of preference based even on electricals may change a bit because of the introduction of an air gap into the design mix.

One last note.... on subjective level... again, this depends on taste and the auditor. Guitar outputs are often (as I understand it from guitar amp manufacturers) but not always... either built on a "lesser" grade of lam like M19 or the designer runs the M6 core significantly harder so that it gets funkier and produces a rich, fat tone with lots of nice harmonics. In hi-fi (for lack of a better term).... we may not be looking directly for funk.... but core material choices may have an impact on subjective sound quality....

>Also I'm not familiar with M4. Does it hold advantages at all levels
>over M6 for output transformers, or is it best used for low power
>designs?

M4 can be run at substantially the same flux density as M6 before saturation sets in. It does not have a marked decrease in maximum flux capacity (relative to M6 grade) as say high (80%) or low (50%) nickels do. the two materials with the highest flux capacities are the superpermendurs (cobalt iron) and 1020 carbon steel. But each of these also have higher (again in a relative sense) losses in watts per pound.

that's it for now...hope I provided some insight....and I hope I've not steered you wrong.... I did this off the top of my head ending at 2:57 AM after I got back from the stock car races wherein my brother (a professional driver) had a horrible outing... but all is well....

Mike LaFevre